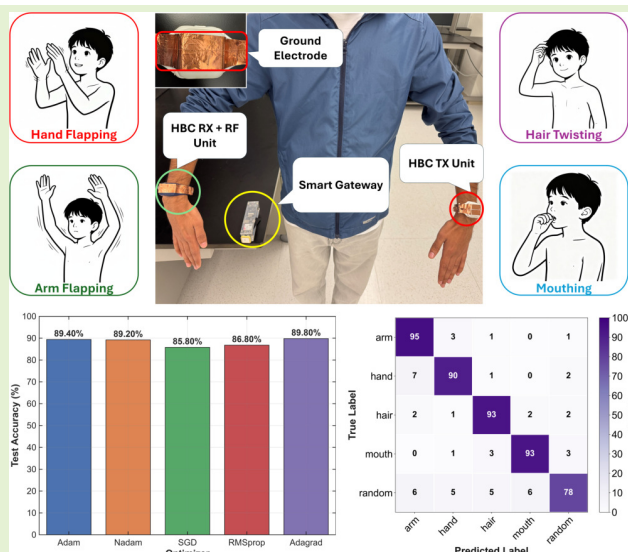


Stereotypic Repetitive Behavior Monitoring Using Human Body Communication

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Abstract—Autism spectrum disorder (ASD) affects about 1 in 127 children globally, with repetitive motor behaviors serving as critical markers for assessment and intervention. While wearable inertial sensors enable continuous behavioral monitoring, practical deployment faces several challenges: radio frequency (RF)-based multisensor systems suffer from high energy consumption, electromagnetic interference, and bulky designs that limit long-term pediatric compliance. This article presents a feasibility study of a human body communication (HBC)-enabled wearable network that addresses these issues through a scalable architecture, where multiple on-body nodes communicate via HBC, and only one node maintains RF capability. HBC leverages the human body as a secure, low-power transmission medium, substantially reducing network energy consumption and interference compared to multi-RF systems. A portable gateway buffers data for context-aware operation and synchronizes with the cloud for clinical visualization. As a proof of concept, a two-node wrist-worn system is implemented to detect upper limb repetitive behaviors, including hand flapping, arm flapping, hair twisting, and mouthing. A custom HBC transmitter (TX) IC, fabricated in TSMC 65-nm technology, achieves core power below 9 μ W, substantially lower than conventional RF solutions. Experiments confirm that HBC preserves accelerometer signal fidelity comparable to direct RF transmission. For automated behavior recognition, a 1-D convolutional neural network (CNN) achieved 91% accuracy on single-subject data and 89.5% on multisubject data at 26 Hz. On 4-Hz data augmented with synthetic sequences from a conditional Wasserstein generative adversarial network (GAN) with gradient penalty, accuracy reached 89.8%. These results highlight HBC-based multinode sensing with deep learning as a scalable pathway for ASD behavioral monitoring.

Index Terms—Accelerometer, autism spectrum disorder (ASD), generative adversarial network (GAN), human body communication (HBC), machine learning (ML).



I. INTRODUCTION

AUTISM spectrum disorder (ASD) is a developmental condition that influences how individuals communicate, perceive sensory information, and manage their behavior. It is characterized by persistent challenges in social interaction and communication, along with repetitive behaviors and

restricted interests [1]. The clinical presentation varies widely across individuals, reflecting the heterogeneity of the spectrum. Recent epidemiological studies show a notable global increase in ASD prevalence, with approximately 1 in 31 children (3.2%) aged eight years in the United States identified with ASD in 2022 compared to 1 in 150 in 2000 [2]. Globally, about 1 in 127 individuals were affected in 2021, with males being diagnosed 3.4 times more frequently than females [3], [4]. This increasing prevalence underscores the need for scalable, objective, and continuous behavioral monitoring strategies that support early detection and tailored intervention. Early identification enables timely access to specialized developmental services and can substantially enhance long-term functional outcomes [5].

Among the core behavioral markers of ASD are repetitive motor and sensory behaviors, collectively referred to as self-stimulatory behaviors or stimming, which serve essential functions in sensory self-regulation and emotional control. These behaviors include hand or arm flapping, rocking, hair twisting, and object mouthing, with their frequency and inten-

Received 12 May 2026; accepted 10 June 2026. Date of publication 24 June 2026; date of current version 17 August 2026. This work was supported in part by the King Abdullah University of Science and Technology (KAUST) Center of Excellence for Smart Health under Award 5932, in part by NEOM under Award 4819, and in part by the Office of President and Vice President of Research at KAUST under Award REP/1/6565-01-01. The associate editor coordinating the review of this article and approving it for publication was Prof. Bin Gao. (Corresponding author: Rajat Kumar.)

This work involved human subjects in its research. Approval of all ethical and experimental procedures and protocols was granted by the Institutional Biosafety and Bioethics Committee (IBEC) under Application No. 20IBEC30.

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Digital Object Identifier 10.1109/JSEN.2026.3705190

sity varying across individuals and contexts. In severe cases, such behaviors may manifest as self-injurious actions (e.g., self-biting or head banging), which can cause physical harm. Accurate detection and quantification of these behaviors are therefore vital for clinical assessment, progress tracking, and intervention planning [6], [7], [8].

This study focuses on upper limb repetitive movements, such as hand flapping, arm flapping, hair twisting, and mouthing, which represent some of the most prevalent and clinically informative behavioral signatures in ASD [6], [9]. These actions are readily observable in natural environments and well-suited for objective analysis through wearable motion sensors. Hand flapping typically involves rapid wrist-originated oscillations, whereas arm flapping originates from the elbow or shoulder [9]. Clinically, the behavior becomes a concern when it persists beyond the typical developmental period, increases in frequency, or interferes with social interaction and daily functioning. In some cases, strong or uncontrolled flapping can also result in self-injury due to accidental contact with nearby objects. Repetitive hair twisting or pulling provides tactile sensory feedback but may result in scalp damage [10], [11]. Similarly, mouthing behaviors, placing nonfood objects in the mouth beyond early developmental stages, pose hygiene and safety risks [12]. Quantitatively analyzing these upper limb behaviors can thus yield valuable biomarkers for ASD screening and behavioral profiling.

Recent advancements in wearable sensing modalities have enabled quantitative studies of motor characteristics in ASD. Inertial sensors, when integrated into wearable systems or smart devices, have demonstrated high discriminatory power in identifying autism-specific motion profiles, with reported classification accuracies exceeding 90% [13], [14]. Beyond motion tracking, wearable devices that measure physiological signals such as heart rate variability or electrodermal activity have facilitated real-time assessment of emotional responses including anxiety or agitation [15]. Computer vision and video-based behavioral analytics have also gained traction; however, their dependence on controlled environments and computational overhead limits their applicability in naturalistic, day-to-day settings [16], [17].

Despite these technological advancements, existing sensor-based ASD monitoring systems face several practical limitations. Multisensor platforms relying on radio frequency (RF) communication suffer from interference in congested frequency bands (e.g., 2.4-GHz ISM band), body shadowing that attenuates signals between on-body nodes, and signal degradation from body movements and antenna misalignment. High power consumption further restricts continuous operation, while bulky device designs and multiple transmitters (TXs) reduce comfort and compliance, particularly in children with sensory sensitivities [18], [19]. In addition, existing sensor- and camera-based technologies can widen healthcare disparities since they typically require expensive hardware, specialized expertise, and controlled environments, which are often unavailable in resource-constrained settings [20]. Furthermore, the presence of visible monitoring devices may alter natural behavior by inducing self-awareness, thereby

diminishing ecological validity and underscoring the need for unobtrusive, context-aware sensing systems [21], [22].

To overcome these limitations, this work introduces a human body communication (HBC)-enabled wearable sensor network for efficient and unobtrusive monitoring of ASD-related repetitive behaviors. HBC employs the human body as a communication medium for intrabody networking, offering significantly lower power consumption and minimized electromagnetic interference relative to traditional RF technologies. While HBC technology has been extensively validated in prior body-area network research, this work focuses on its system-level integration for behavioral monitoring applications, demonstrating end-to-end signal fidelity and classification performance. The proposed architecture employs a scalable multinode design where multiple on-body sensor nodes communicate via HBC channels, while only a single node maintains RF connectivity to relay aggregated data to external devices or cloud infrastructure. This approach enables flexible sensor placement across the body while restricting high-power RF transmission to a single point, thereby minimizing interference, reducing overall network energy consumption, and simplifying integration with existing wireless ecosystems. As a proof-of-concept case study, we implement a two-node wrist-worn system for monitoring upper limb repetitive behaviors in ASD. In this configuration, one wrist sensor transmits motion data over HBC to the second wrist sensor, which aggregates bilateral measurements and maintains RF connectivity to a portable hub. The hub, carried in a pocket or bag, buffers data for context-aware processing (incorporating location, time, and environmental context) and synchronizes with cloud-based analytics platforms to support clinical assessment. This bilateral sensing configuration enables continuous monitoring of hand flapping, arm flapping, hair twisting, and mouthing behaviors while demonstrating the scalability and energy efficiency of the HBC-based body-area network paradigm. The main contributions of this work are as follows.

- 1) A compact, multimodal wearable sensing platform for real-time monitoring and classification of ASD-related upper limb repetitive behaviors, supported by a context-aware gateway for local buffering/processing and cloud synchronization for clinician review.
- 2) An HBC-enabled architecture using capacitive coupling to replace most RF links, improving robustness and reducing system energy compared with RF-only wearable networks.
- 3) A custom ultralow-power HBC TX integrated circuit (IC) supporting on-off keying (OOK) modulation.
- 4) An optimized neural-network-based classifier achieving 91% accuracy on single-subject data, 89.5% on multi-subject data, and 89.8% on low-sampling-rate data with synthetic data augmentation.

The remainder of this article is organized as follows. Section II describes the system architecture and data acquisition. Section III presents hardware implementation and characterization. Section IV evaluates classification performance. Section V discusses limitations and future work.

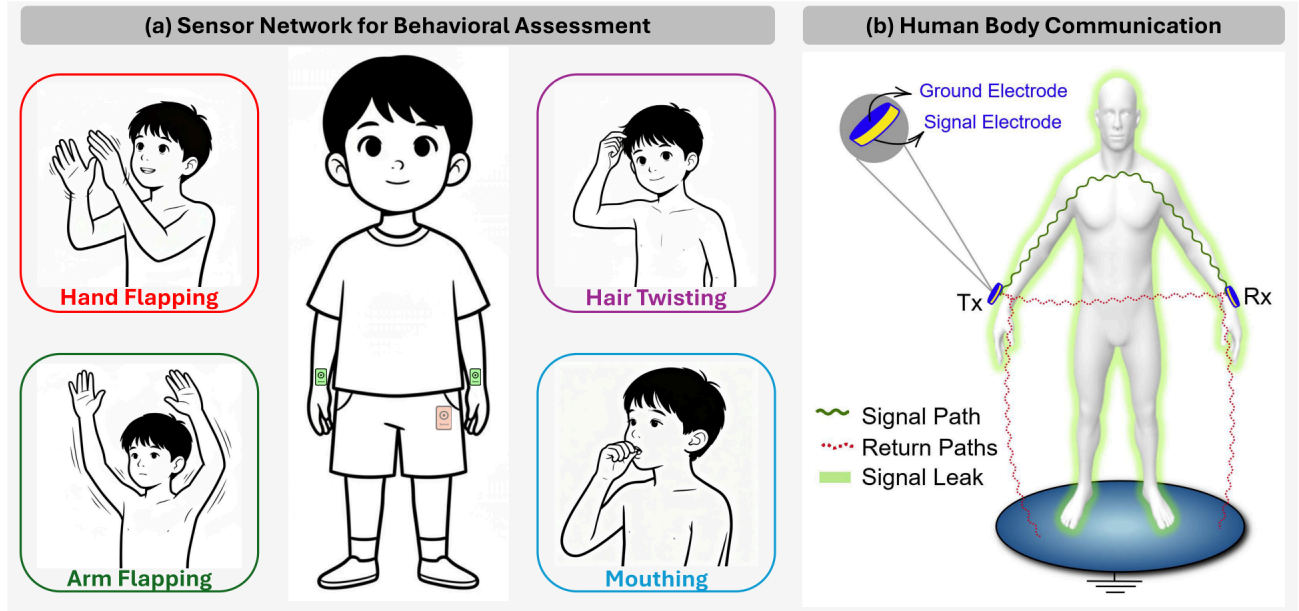


Fig. 1. (a) Illustration of a child equipped with wrist sensors and a pocket hub, highlighting the repetitive behaviors: hand flapping, arm flapping, hair twisting, and mouthing. (b) Schematic of capacitive coupling HBC, showing signal electrodes connected to the skin and ground electrodes floating in air.

Finally, Section VI offers concluding remarks summarizing the key findings of this study.

II. SYSTEM ARCHITECTURE AND DATA ACQUISITION

A. System Overview

Continuous monitoring of repetitive behaviors in individuals with ASD is important for early detection, longitudinal tracking, and timely therapeutic intervention within sensor-informatics-driven healthcare workflows. The proposed HBC-enabled wearable platform provides an integrated framework for capturing upper limb movement patterns associated with common repetitive behaviors, including arm flapping, hand flapping (including rapid object hitting), hair twisting (including hair pulling), and mouthing. As illustrated in Fig. 1(a), motion sensors are worn on both wrists to collect kinematic signals, while a compact hub carried in a pocket or bag aggregates sensor data, integrates contextual information, and transmits the results to a cloud-based analytics platform. The cloud system performs algorithmic analysis to derive quantitative behavioral metrics, which are then displayed on the clinician's interface for remote review and assessment.

The key innovation of this monitoring framework lies in its use of HBC for intrabody data transfer between the two wrist sensors. Rather than having multiple sensors each transmitting via power-hungry RF signals, the proposed system uses the human body itself as a communication channel. The proposed design employs capacitive-coupling HBC, where the TX and receiver (RX) signal electrodes are in contact with the skin and the ground electrodes remain floating, as shown in Fig. 1(b). In this mode, the electric field generated at the TX induces a displacement current that propagates through the body and closes its return path through parasitic capacitance to the surrounding

environment [23]. Recent progress in HBC transceivers has substantially improved link reliability and energy efficiency for wearable sensing. Compared to conventional RF links, HBC can reduce energy consumption by up to two orders of magnitude, minimize device size, and limit signal leakage. Moreover, operation in the electroquasistatic regime provides a leakage detection range reported to be up to 32 times shorter than typical RF systems, which improves privacy and makes HBC well-suited for wearable sensing applications [24].

The complete system comprises three principal components operating jointly: 1) an HBC TX wrist unit equipped with an accelerometer and custom HBC IC for intrabody data transmission; 2) an HBC RX wrist unit that combines local sensing with RF communication capabilities for bilateral data aggregation and cloud interfacing; and 3) a smart gateway (hub) that performs final multisensor fusion, enriches data with contextual information (e.g., location and time-of-day), and manages cloud synchronization via cellular connectivity. The system architecture strategically separates low-power on-body sensing from high-power wide-area communication: the TX wrist sensor transmits its motion data through the body to the RX wrist sensor, which aggregates data from both wrists and maintains the only RF link to the portable hub. Fig. 2 illustrates this data flow from sensors through the body, to the hub, and ultimately to the cloud for clinical analysis.

The proposed multisensor HBC system provides three main advantages aligned with wearable sensor-informatics requirements for longitudinal monitoring as follows.

- 1) *Extended Battery Duration:* The low-power nature of HBC significantly prolongs sensor runtime, allowing smaller batteries and improved comfort for long-term use.

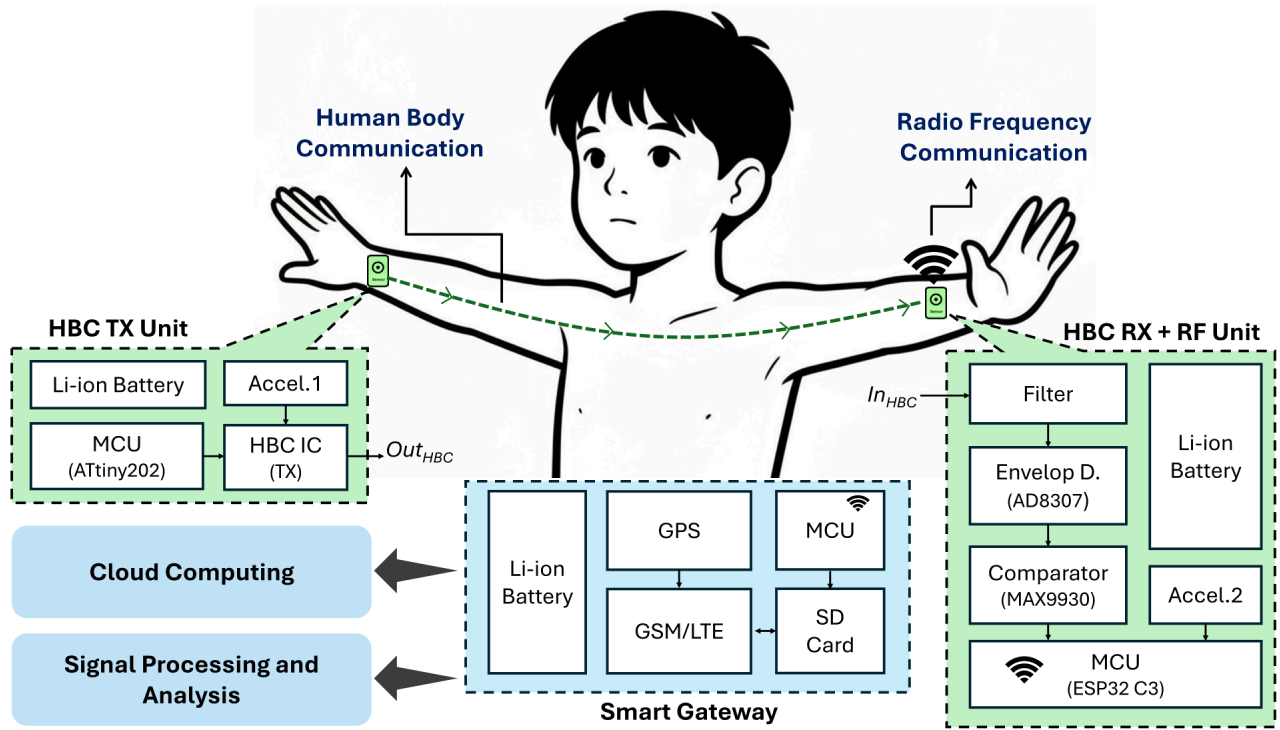


Fig. 2. Block-level overview of the proposed hardware platform. The system comprises an HBC TX wrist unit, an HBC RX + RF wrist unit, and a smart gateway that aggregates multisensor data, adds contextual information, and relays processed features to the cloud for analytics.

- 2) *Reduced Interference*: Only the hub unit communicates with the cloud via RF, eliminating cross-interference between multiple sensors on the body.
- 3) *Contextual Information Integration*: The hub synchronizes and fuses sensor and environmental data, enabling clinicians to identify patterns, triggers, and behavioral contexts more effectively.

B. Data Acquisition

The data acquisition protocol captures representative inertial motion signals associated with distinct repetitive behaviors performed at varying intensities, postures, and movement frequencies. Pitch and roll components are recorded for each activity to provide detailed kinematic information suitable for subsequent feature extraction and temporal pattern analysis. Each activity instance is recorded for 5 s to ensure sufficient temporal coverage for dynamic characterization of the underlying motion patterns.

Sampling-rate selection is guided by prior kinematic studies of hand flapping, which is the fastest behavior among the considered classes. Reported interflap periods in autistic individuals indicate a flap frequency of approximately $f_{\text{flap}} \in [0.67, 3.33]$ flaps/s [25]. Because each flap typically yields two prominent acceleration peaks (upstroke and downstroke), the peak occurrence rate can reach $2f_{\text{flap,max}} \approx 6.67$ Hz. The Nyquist criterion therefore implies a minimum sampling frequency of about 13.3 Hz. A separate study in which multiple participants mimicked the behavior found dominant motion content in the 8–13-Hz band [26]. Based on these observations, a higher rate configuration of approximately

26 Hz (about 130 samples per 5-s window) is selected to satisfy the Nyquist–Shannon criterion with margin and to better capture transient peaks, harmonic components, and subtle transitions in the derived pitch–roll trajectories.

While such a high-rate configuration is essential for accurately characterizing rapid and transient behaviors like hand flapping, not all behaviors demand this temporal resolution. Slower repetitive actions such as hair twisting and mouthing exhibit much lower motion dynamics and can be adequately captured at reduced rates. Moreover, the reported flap-frequency range indicates considerable variability in the number of flaps that may occur within a fixed analysis window. To enable energy-efficient wearable monitoring by reducing both sensor sampling activity and the volume of wireless data transmitted, a low-rate configuration of approximately 4 Hz (about 20 samples per 5-s window) is also evaluated.

Accordingly, inertial data are recorded from a single human subject wearing two wrist-mounted accelerometers, one on each wrist. The right-wrist node operates at a high sampling rate of approximately 26 Hz, yielding about 130 samples per 5-s window, whereas the left-wrist node uses a reduced sampling rate of about 4 Hz, producing roughly 20 samples in the same period. For each predefined repetitive behavior, 100 sequences are collected under controlled conditions to minimize sensor drift and intertrial variability. This procedure yields 400 labeled motion segments evenly distributed across four activity classes. An additional 100 random-motion sequences corresponding to spontaneous, unstructured movements are also recorded to enhance classifier robustness. The resulting single-subject datasets are designated as Dataset-S01_RW (right wrist, high sampling rate, and 500 samples)

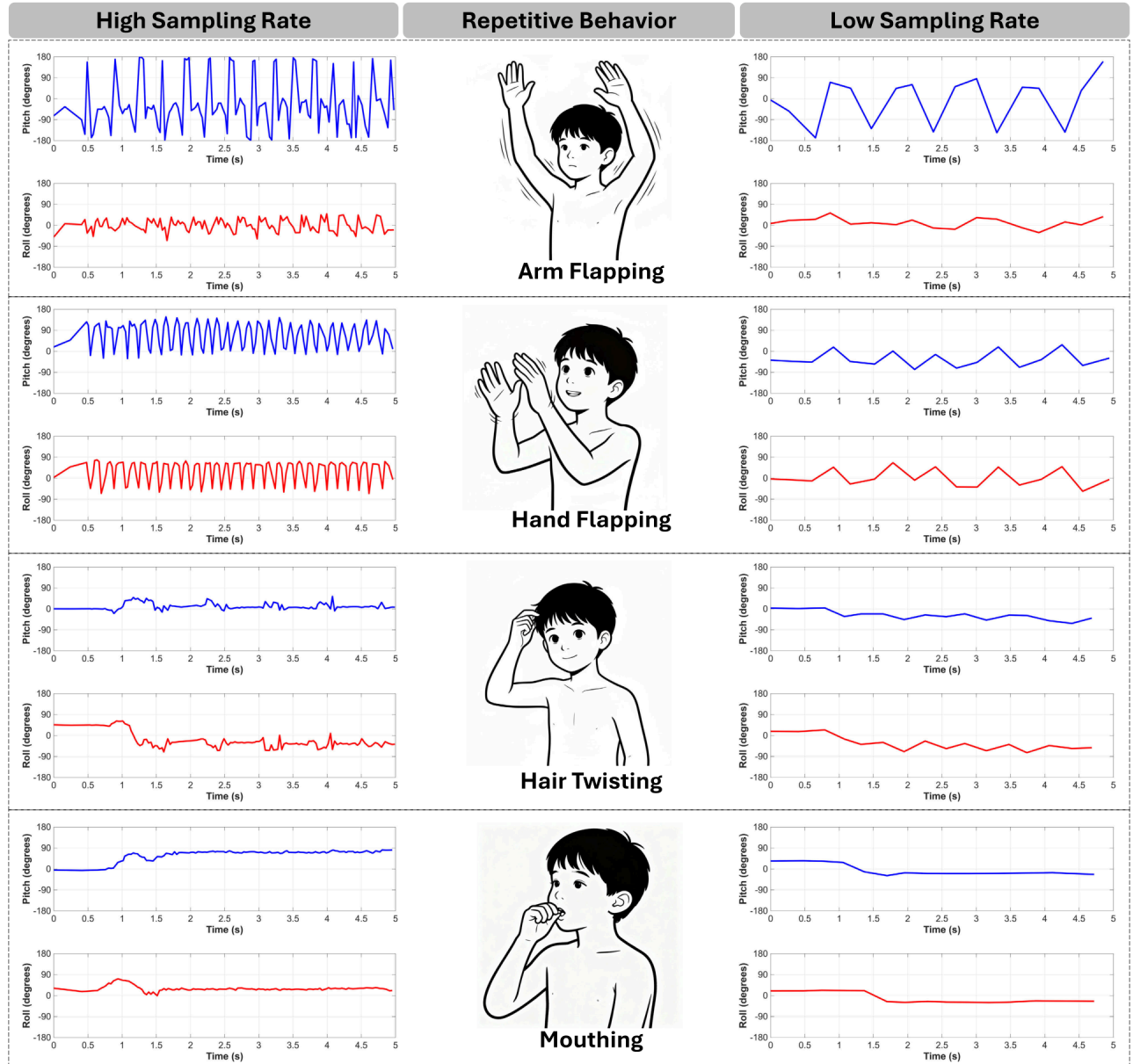


Fig. 3. Representative pitch and roll time-series segments for four ASD-related behaviors (arm flapping, hand flapping, hair twisting, and mouthing) recorded from the wrist-worn inertial sensors, illustrating characteristic motion patterns and interclass separability.

and Dataset-S01_LW (left wrist, low sampling rate, and 500 samples).

To expand the dataset and enable population-level generalization, the same recording protocol is then extended to a group of 20 additional participants using the high-rate configuration. Each participant performs every predefined behavior 10 times, resulting in 200 samples per activity class and a total of 800 labeled motion sequences across the four classes. Random-motion data representing spontaneous, unstructured hand and wrist movements are also collected to match the total number of samples across all activity classes, ensuring a balanced dataset composition. Depending on the specific analysis requirements, additional random-motion sequences are recorded and selectively evaluated to support experiments on unbalanced class distributions and model robustness under variable sample ratios. The participant cohort includes both

male and female users aged between 23 and 35 years and represents diverse geographical regions to increase variability in natural motion patterns. All participant information is anonymized, and informed consent is obtained from each individual in accordance with institutional ethical guidelines. The combined multiuser dataset, referred to as Dataset-SXX_RW, preserves identical recording parameters, segment length, and annotation structure as in the single-subject dataset, thereby facilitating cross-user validation and multisubject model generalization.

The labeling of the behaviors is performed with the advice of clinicians working in the field to ensure accurate and contextually valid annotations. Representative time-series waveforms for each activity class are shown in Fig. 3, highlighting characteristic motion dynamics and separability across behaviors.

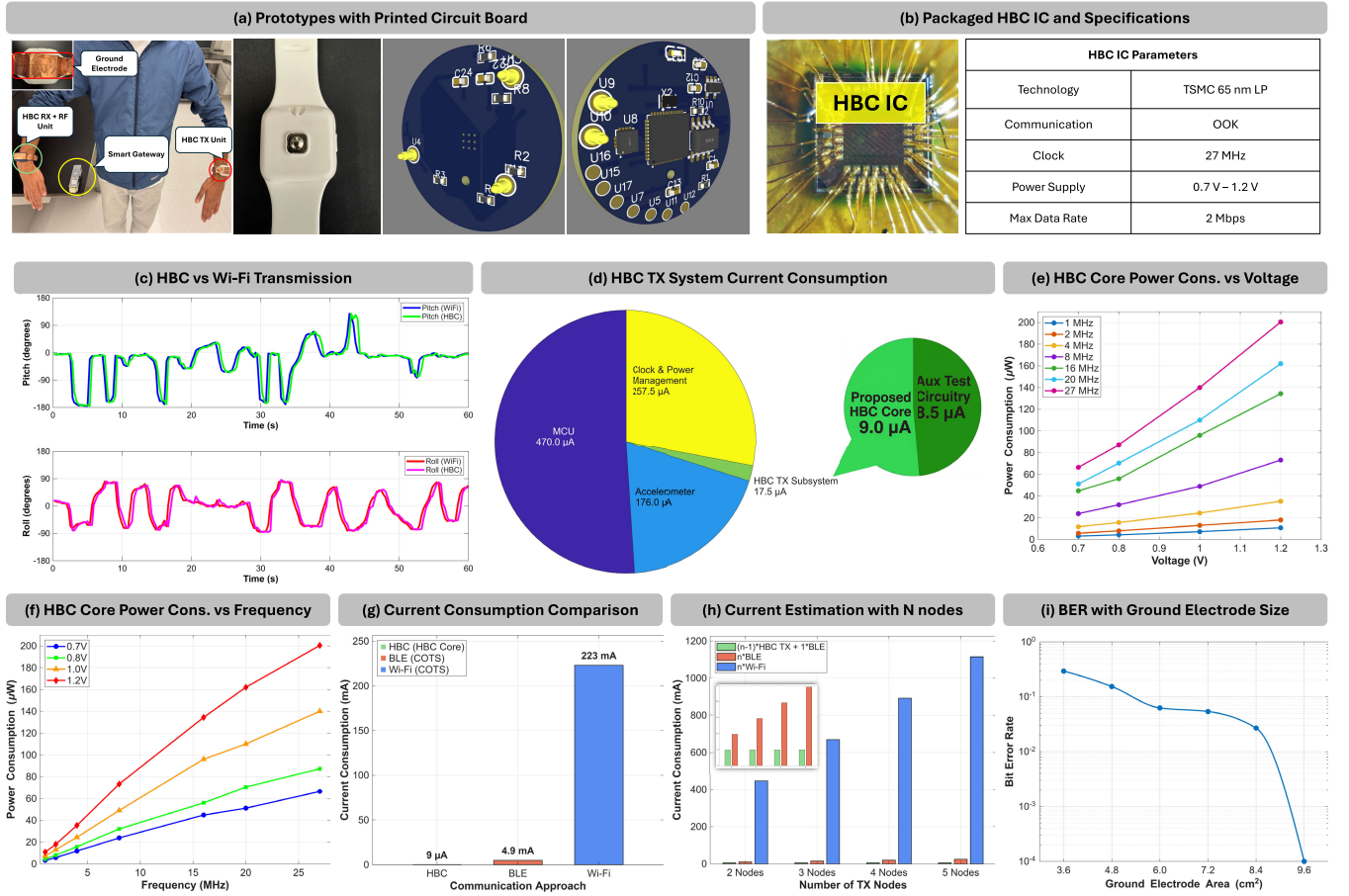


Fig. 4. (a) Complete system implementation showing the assembled prototype and miniaturized PCB modules. (b) Packaged 65-nm CMOS HBC IC and key specifications. (c) Pitch and roll trajectories measured locally and received over HBC and RF links, demonstrating waveform fidelity after transmission. (d) Current consumption breakdown of the HBC TX node across major PCB subsystems. (e) Measured HBC core power as a function of operating frequency. (f) Measured HBC core power as a function of supply voltage. (g) Power comparison between HBC, BLE, and Wi-Fi transmissions under the same platform conditions. (h) Projected total current consumption versus the number of HBC sensor nodes, highlighting scalability benefits relative to RF-based links. (i) Characterization of HBC channel performance using BER as a function of ground electrode size.

III. HARDWARE IMPLEMENTATION AND ANALYSIS

This section describes the hardware implementation of the proposed system, covering platform design, communication protocol and signal processing, and energy efficiency characterization.

A. Hardware Platform Design

The proposed wearable sensing system is realized through two wrist-worn prototypes implemented in compact wristband form factors. The first wrist unit integrates all key components—including a custom HBC TX IC, accelerometer (BMI160), and microcontroller (ATtiny202)—on miniaturized PCBs housed within the wristband, while a rear electrode maintains direct skin contact to ensure robust capacitive HBC coupling. The complete prototype and its PCB assemblies are shown in Fig. 4(a). The HBC TX IC is fabricated in a TSMC 65-nm CMOS technology node, with the digital implementation carried out using Cadence standard design tools. The resulting design occupies 0.1 mm² and integrates approximately 18-k logic gates. The packaged HBC IC and its key specifications are illustrated in Fig. 4(b). For intrabody communication, the IC supports OOK modulation, which is

sufficient to reliably convey inertial payloads through the human body under the proposed capacitive-coupling configuration. The HBC IC incorporates a UART interface that enables flexible configuration of key transmission parameters, including data rate and carrier frequency selection. An ATtiny202 microcontroller unit (MCU) manages sensor communication, data acquisition from the accelerometer, and configuration of the HBC IC for packet transmission over the body channel.

The RX unit, worn on the opposite wrist, integrates three primary components: an HBC RX chain, a local accelerometer, and an RF communication module. The HBC RX chain is implemented using discrete commercial off-the-shelf (COTS) components to validate RX functionality in this proof-of-concept system, enabling rapid prototyping, flexible testing, and iterative optimization. The chain comprises a narrowband band-pass filter for signal isolation, an envelope detector (AD8307), and a comparator (MAX9930) to recover the transmitted waveform from the contralateral wrist. A local accelerometer (BMI160) captures motion data from the wrist wearing the RX unit, enabling bilateral sensing. An ESP32-C3 microcontroller processes the received HBC data, synchronizes it with local accelerometer measurements, and manages wire-

less connectivity via BLE or Wi-Fi for cloud synchronization. The RX unit is designed for integration into a smartwatch platform with substantially larger battery capacity, allowing it to accommodate the RX and RF circuitry without compromising wearability. While full ASIC integration of the RX remains beyond the scope of the current phase, the future RX-ASIC implementation would follow a standard custom-IC flow, beginning with system-level specification of sensitivity, bandwidth, dynamic range, and power budget, followed by transistor-level design of the low-noise front-end, bandpass filtering, envelope detection, comparator, and baseband/output-conditioning stages. The design would then proceed through schematic validation, prelayout simulation, layout implementation with careful device matching and parasitic control, postlayout extraction, Monte Carlo and corner verification, and final tape-out. This discrete COTS-based implementation serves as a proof-of-concept that validates the reception principle and provides an engineering baseline for a compact, low-power, single-chip RX suitable for smartwatch platforms and other space-constrained wearable devices.

The smart gateway functions as a central aggregation node, collecting data streams from all wearable devices through RF communication and supporting both local storage and cloud synchronization. It integrates auxiliary sensors, such as a global positioning system (GPS) module, to capture contextual and environmental information, and includes an onboard SD card for extended local logging in offline or low-connectivity environments. In such cases, data are stored locally on the gateway and can be uploaded to the cloud once network access is restored. A cellular interface, such as global system for mobile communications (GSM) or long-term evolution (LTE), enables remote data transmission and analytics when connectivity is available. Since the gateway is not worn on the body, it can accommodate a higher capacity battery and additional processing and communication hardware, supporting prolonged operation and autonomous field deployment. The gateway architecture is also compatible with future secure implementations that incorporate encrypted transfer, authentication, and anonymized record handling to better protect user privacy.

B. Data Communication and Preprocessing

Reliable data communication over the HBC link is achieved using a structured packet frame that ensures synchronization and payload integrity. Each packet comprises 8 start bits, 4 length bits, a 32-bit payload, and an 8-bit cyclic redundancy check (CRC), where the start bits maintain RX synchronization, the length field specifies the payload size in bytes, and the CRC enables error detection to ensure data integrity. The payload encodes pitch and roll measurements, each quantized to 16 bits and derived from on-board accelerometer readings.

Raw accelerometer outputs are first converted to physical acceleration in units of g using the device sensitivity. For a full-scale range of ± 2 g and a sensitivity of 16384 LSB/g, the acceleration components are computed as

$$a_x = \frac{ax_raw}{16384.0} \quad a_y = \frac{ay_raw}{16384.0} \quad a_z = \frac{az_raw}{16384.0}.$$

The ± 2 g configuration adequately covers typical human motion while maximizing resolution (approximately 0.06 mg/LSB), thereby reducing quantization noise and improving downstream feature extraction for machine learning (ML)-based behavioral classification. Pitch and roll angles in degrees are then obtained using

$$\text{pitch} = \arctan 2(a_y, a_z) \cdot \frac{180.0}{\pi} \quad (1)$$

$$\text{roll} = \arctan 2\left(-a_x, \sqrt{a_y^2 + a_z^2}\right) \cdot \frac{180.0}{\pi}. \quad (2)$$

The HBC link validation focuses on demonstrating end-to-end signal fidelity preservation for the behavioral monitoring application. Fig. 4(c) compares pitch and roll trajectories measured directly from the accelerometer with those received over the HBC and RF links. The close overlap between the HBC and local accelerometer traces indicates preserved waveform fidelity. This alignment confirms that the proposed HBC link provides stable, low-data-rate transmission suitable for continuous motion monitoring in wearable systems.

C. Energy Efficiency and Channel Characterization Analysis

Energy efficiency of the system is evaluated using the custom HBC TX. Since the architecture is designed to support multiple distributed TX nodes with only a single RX node (integrated into a smartwatch platform with larger battery capacity), TX power consumption is the critical factor for system scalability and therefore the focus of this analysis. HBC achieves substantially lower energy consumption than conventional wireless standards such as BLE and Wi-Fi used in similar wearable applications. Fig. 4(d) details the current distribution of the HBC TX system implemented with the custom IC, highlighting that the proposed HBC core itself consumes only $9 \mu\text{A}$, with the remaining current attributed to the MCU, accelerometer, and clock and power management circuitry. A more detailed power characterization of the HBC core is performed across supply voltages and operating frequencies, with results summarized in Fig. 4(e) and (f). The power consumption increases with both voltage and frequency, with the maximum measured value of $200 \mu\text{W}$ at 1.2 V and 27 MHz. Fig. 4(g) compares the TX current consumption of the proposed HBC Tx core with BLE and Wi-Fi using representative values from a COTS platform. HBC draws $9 \mu\text{A}$, versus 4.9 mA for BLE and 223 mA for Wi-Fi, underscoring HBC's suitability for ultralow-power, always-on sensing. The BLE and Wi-Fi currents are reported as typical reference values achievable with compact, low-power implementations, and duty cycling is not considered in this comparison [27], [28].

To evaluate scalability, multinode coordination is estimated using frequency-divided multiple access (FDMA), enabling synchronized operation and sequential acquisition from multiple TX nodes without compromising link robustness. Fig. 4(h) shows the projected total communication current consumption as the number of TX nodes increases. The proposed HBC + BLE approach scales with only a modest increase in total current (dominated by the BLE RX) while

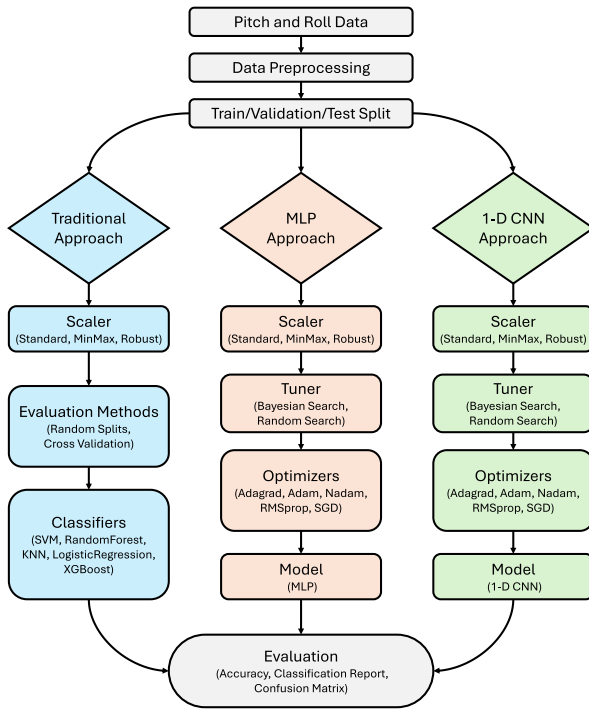


Fig. 5. ML pipeline for behavior classification from wrist-worn inertial data. The flowchart summarizes the stages from signal acquisition and preprocessing, through dataset splitting and feature scaling, to model training, hyperparameter tuning, and final performance evaluation.

remaining well below the consumption of comparable RF-based implementations, even at higher node densities. These results indicate that the proposed HBC platform preserves strong energy efficiency under network scaling, making it a promising candidate for multisensor body-area networks and continuous behavioral monitoring in wearable systems. Notably, this projection accounts only for communication power consumption and does not include the processing power required for sensing or on-node computation.

The channel characterization of HBC, covering bit error rate (BER) across various data rates, power supplies, and communication methods, has been extensively reported in our prior work for different HBC application scenarios [24], [29], [30], [31], [32]. The proposed system is evaluated using stereotypical movement recordings collected from multiple subjects under realistic conditions, including variations in body motion, posture, and skin-contact quality, to reflect practical wearable use. In the context of this work, however, the most relevant channel parameter is the ground electrode size on the wearable device, as it directly governs coupling quality and signal transmission integrity. To this end, BER measurements are conducted in a fixed environment across varying ground electrode areas. The results, plotted in Fig. 4(i), reveal that once the electrode area exceeds 9.6 cm², the BER drops to a significantly low level, establishing a practical design threshold for reliable HBC operation.

IV. BEHAVIOR CLASSIFICATION RESULTS

A. Single-User Classification Performance

The analysis begins by evaluating several ML pipelines for classifying repetitive behaviors from wrist-worn inertial

measurements using the right-wrist dataset (Dataset-S01_RW). Representative pitch-roll time-series samples for each behavioral category are shown in Fig. 3, and the end-to-end processing and evaluation workflow is summarized in Fig. 5. Starting from raw pitch and roll signals, the pipeline standardizes sequence length, applies feature scaling, and trains models under a consistent data-splitting protocol. No threshold-based motion detection is applied; instead, recordings are collected under controlled, activity-driven conditions, and preprocessing is limited to sequence standardization prior to classification. The dataset is then partitioned into training and testing sets using an 80:20 split, with 20% of the training set further reserved for validation to support unbiased hyperparameter tuning and model selection. Partitioning is performed at the sample level using stratified random splitting across the pooled dataset rather than at the subject level. Consequently, samples from the same participant may appear in both the training and test sets, and the reported performance reflects within-distribution generalization.

1) Traditional ML Approach: Pitch and roll sequences from the dataset are reshaped into fixed-length feature vectors of $T \times 2$ dimensions to enable the application of traditional ML classifiers. Five classifiers—support vector machine (SVM), random forest (RForest), k -nearest neighbors (KNNs), logistic regression (LogReg), and XGBoost—are evaluated alongside three feature scaling methods: StandardScaler, MinMaxScaler, and RobustScaler. Classifier hyperparameters are fixed based on initial tuning experiments, and StandardScaler is selected for subsequent analyses following comparative evaluation.

To assess performance and variability, two evaluation schemes are used, as illustrated in Fig. 6(a) and (b). The first scheme uses repeated random splits (50 iterations) to estimate mean accuracy while capturing split-induced variance. The second scheme uses repeated stratified fivefold cross-validation (ten repetitions) to provide a more statistically reliable estimate by ensuring balanced testing across stratified folds. As shown in Fig. 6(a), average test accuracies from repeated random splits indicate that RForest achieves the highest performance at approximately 82%. A consistent trend is observed in the stratified K-fold evaluation, as illustrated in Fig. 6(b), where RForest again attains 82%. However, these results remain sub-optimal for the behavioral classification task, which requires higher accuracy for practical deployment. Consequently, more advanced approaches that explicitly model temporal dependencies and complex motion patterns in the kinematic sequences are required.

2) MLP Approach: To overcome the limited representational capacity of traditional classifiers, a multilayer perceptron (MLP) is trained on the same flattened pitch-roll representation and evaluated under identical preprocessing options, including StandardScaler, MinMaxScaler, and RobustScaler. The network takes a $T \times 2$ input and consists of two fully connected (FC) blocks with ReLU activations: the first uses 64–256 units and the second uses 32–128 units, each followed by batch normalization and dropout (0.2–0.5), and includes L1/L2 regularization. A final softmax layer produces predictions over five classes. Model selection relies on hyperparameter tuning with two strategies: Bayesian search and

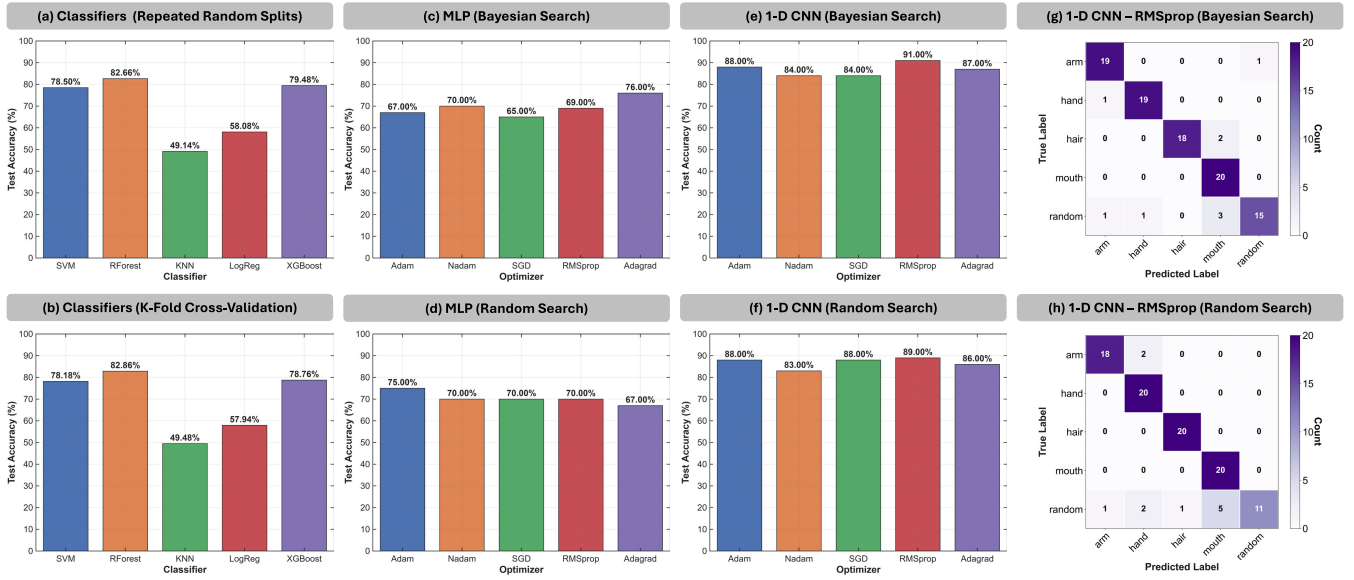


Fig. 6. Classification performance on the right-wrist dataset (Dataset-S01.RW). (a) Average test accuracies of traditional ML classifiers under repeated random splits. (b) Average test accuracies of traditional ML classifiers under repeated stratified fivefold cross-validation. (c) MLP test accuracies for different optimizers using Bayesian hyperparameter search. (d) MLP test accuracies using random search. (e) 1-D CNN test accuracies with Bayesian search. (f) 1-D CNN test accuracies with random search. (g) Confusion matrix of the best-performing 1-D CNN model obtained using Bayesian search. (h) Confusion matrix of the best-performing 1-D CNN model obtained using random search.

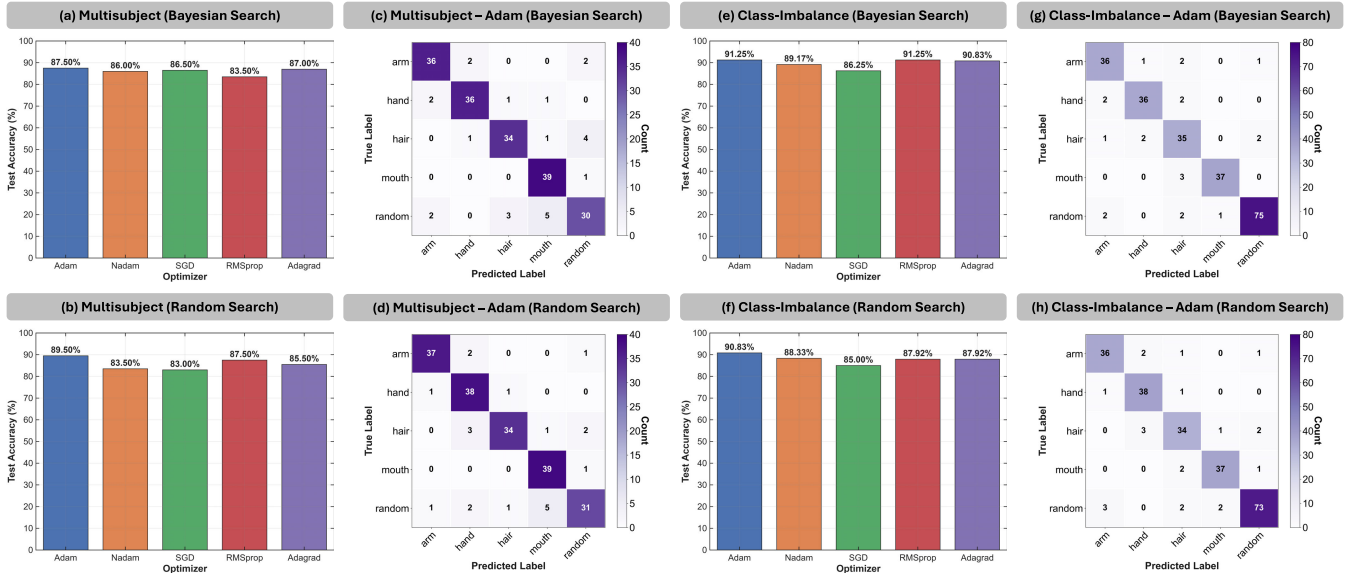


Fig. 7. Performance of the proposed 1-D CNN under multisubject and class-imbalanced conditions. (a) and (b) Hyperparameter optimization results for the multisubject dataset using Bayesian optimization and random search, respectively. (c) and (d) Corresponding confusion matrices for the best-performing optimizer, Adam, under Bayesian optimization and random search. (e) and (f) Hyperparameter optimization results for the class-imbalanced dataset using Bayesian optimization and random search, respectively. (g) and (h) Corresponding confusion matrices for the best-performing optimizer, Adam, under Bayesian optimization and random search.

random search (100 trials each). The search space spans the optimizer (Adam, Nadam, SGD, RMSprop, and Adagrad) and the learning rate in the range 10^{-4} – 10^{-2} . Based on preliminary experiments, StandardScaler and ReLU are fixed for subsequent runs to reduce confounding effects from preprocessing and activation choice.

Fig. 6(c) and (d) summarizes the hyperparameter-tuning outcomes. As illustrated in Fig. 6(c), Bayesian search yields the best test accuracy with Adagrad (76%), followed by Nadam (70%), RMSprop (69%), Adam (67%), and SGD

(65%). Fig. 6(d) shows that random search yields the best test accuracy with Adam (75%), while Nadam, RMSprop, and SGD each reach 70%, and Adagrad attains 67%. Despite improved flexibility and automatic feature learning, the MLP processes each sequence as a fixed-length vector and therefore does not explicitly model temporal structure or local motion patterns in the pitch-roll trajectories. As a result, the best MLP performance (76% with Bayesian Adagrad and 75% with random Adam) does not consistently exceed the strongest traditional baseline (RForest at 82%), suggesting

that feedforward architectures alone are insufficient for robust inertial-signal behavioral classification.

3) 1-D CNN Approach: To exploit the temporal structure of the inertial signals, the classification pipeline replaces FC models with a 1-D convolutional neural network (CNN) that operates directly on the pitch-roll sequences. The input to the network has shape $T \times 2$, and the data preprocessing uses StandardScaler fit only on the training split to avoid data leakage. Based on preliminary comparisons, StandardScaler with ReLU activation provides the most stable training behavior and is used in the final experiments.

The 1-D CNN consists of two convolutional blocks followed by a compact classifier head. Each block applies Conv1D with {16–64} filters in the first layer and {32–128} filters in the second layer, kernel sizes in {3,5}, batch normalization, and max-pooling (pool size 2), enabling the network to learn local temporal patterns while progressively reducing the sequence resolution. The feature extractor is followed by a Flatten layer and a dense layer (32–128 units, ReLU) with dropout (0.2–0.5) and L1/L2 regularization, and a final softmax layer for five-class classification. Hyperparameter tuning uses Bayesian search and random search (100 trials each) over the convolutional filter counts, kernel sizes, dense units, dropout rate, and learning rate (10^{-4} – 10^{-2}), while also comparing optimizers (Adam, Nadam, SGD, RMSprop, and Adagrad).

Fig. 6(e) shows that Bayesian search-based optimization achieves the best test accuracy of 91% with RMSprop, while the remaining optimizers also reach strong performance (Adam: 88%, Adagrad: 87%, Nadam: 84%, and SGD: 84%). Under random search, as shown in Fig. 6(f), RMSprop again provides the best result (89%), with Adam and SGD achieving 88%, Adagrad reaching 86%, and Nadam obtaining 83%. Fig. 6(g) and (h) further presents the confusion matrices corresponding to the best-performing 1-D CNN models obtained using Bayesian and random search, respectively, showing strong diagonal dominance that indicates high class-wise accuracy and limited misclassification across behaviors. Overall, the 1-D CNN consistently improves upon traditional ML and MLP baselines by learning discriminative temporal features directly from the sequential pitch-roll measurements.

B. 1-D CNN Robustness Analysis

1) Multisubject Performance: To assess robustness against intersubject variability, the proposed 1-D CNN is evaluated on the multisubject dataset, Dataset-SXX_RW, which aggregates recordings from 20 subjects with diverse motion kinematics and behavior execution styles. The same preprocessing pipeline used in the single-subject experiments is applied, including feature standardization using StandardScaler fit only on the training split and sequence-length normalization to $T = 130$ timesteps. The dataset contains 200 samples for each of the five behavioral classes, yielding 640 training samples, 160 validation samples, and 200 test samples.

Hyperparameter tuning is performed using the same Bayesian optimization and random search protocols as in the single-subject case, with 100 trials for each method over the same convolutional and optimization search spaces.

Fig. 7(a) and (b) summarizes the optimization results for the multisubject dataset. Bayesian optimization identifies Adam as the best-performing optimizer, achieving 87.5% test accuracy, followed by Adagrad (87.0%), Nadam (86.0%), SGD (86.5%), and RMSprop (83.5%). Random search yields a higher peak accuracy of 89.5% with Adam, while also producing strong results with RMSprop (87.5%), Adagrad (85.5%), Nadam (83.5%), and SGD (83.0%). Overall, the performance of all optimizers remains within a relatively narrow range, indicating stable behavior across tuning strategies.

Fig. 7(c) and (d) shows the confusion matrices corresponding to the best optimizer, Adam, under Bayesian optimization and random search, respectively. In both cases, the matrices reveal strong diagonal dominance, indicating reliable class discrimination across the five behaviors. Minor confusions occur between some behavioral classes and the random-motion category, but the overall classification pattern remains consistent. These results demonstrate that the proposed 1-D CNN generalizes effectively across subjects and retains strong performance even under intersubject variability.

2) Class-Imbalanced Performance: The proposed 1-D CNN is also evaluated under a class-imbalanced setting to better reflect practical monitoring conditions, where random or unstructured movements may occur more frequently than predefined behaviors. To simulate this scenario, the multiuser dataset is augmented with 200 additional random-motion samples, increasing the size of the random class to 400 while keeping the arm, hand, hair, and mouth classes at 200 samples each. The same preprocessing procedure is maintained, including feature standardization and sequence-length normalization to $T = 130$ timesteps. After splitting, the resulting dataset contains 768 training samples, 192 validation samples, and 240 test samples.

Hyperparameter tuning is conducted using Bayesian optimization and random search with 100 trials each over the same search space. Fig. 7(e) and (f) reports the corresponding optimization outcomes. Bayesian optimization achieves a peak test accuracy of 91.25% with both Adam and RMSprop, followed by Adagrad at 90.83%, Nadam at 89.17%, and SGD at 86.25%. Random search reaches 90.83% with Adam, while also producing strong performance with Nadam (88.33%), RMSprop and Adagrad (87.92% each), and SGD (85.00%).

Fig. 7(g) and (h) presents the confusion matrices for the best optimizer under Bayesian optimization and random search, respectively. In both cases, the random-motion class is classified with high reliability, while only limited confusion is observed with a few behavioral classes. The slightly higher accuracy in this setting is mainly attributed to the increased number of random-motion samples, which provides the model with a clearer and more diverse representation of nongesture movements during training. As a result, the classifier becomes better at separating random motions from intentional behaviors, thereby improving the overall recognition performance in this imbalanced scenario.

3) Low-Sampling-Rate Performance: To examine the trade-off between temporal resolution and deployment efficiency, the proposed 1-D CNN is evaluated under a low-sampling-rate setting for both the single-user and multiuser datasets.

In this configuration, each 5-s window is downsampled to 20 timesteps while preserving the same preprocessing pipeline and model architecture used in the full-resolution analysis. The 4-Hz sampling rate falls below the Nyquist rate for hand flapping and is adopted as a deliberate energy-efficiency tradeoff; the reported results therefore represent a lower bound on accuracy under power- and bandwidth-constrained deployment. This low-rate setting is particularly relevant for wearable applications, where reduced sampling directly lowers sensing overhead, communication load, and energy consumption in battery-powered body-area systems.

For the single-user dataset, Bayesian optimization achieves a best test accuracy of 88% with Adam. For the multiuser dataset, the same low-sampling-rate configuration is applied, and the best test accuracy is 84% using random search with Adam. Although these results are lower than those obtained at the higher temporal resolution, they confirm that the proposed 1-D CNN still captures the most informative motion patterns even when the number of observations per sequence is substantially reduced. The obtained accuracies are still higher than those achieved by the other approaches at high temporal resolution, indicating that the model remains functional in bandwidth- and power-constrained scenarios where continuous high-rate sensing may not be practical.

At the same time, the reduction in temporal resolution introduces a clear drop in accuracy, indicating that some discriminative motion details are lost when the sequence is compressed. This observation motivates further investigation of low-rate operation, particularly to improve generalization under reduced temporal resolution. Therefore, more diverse training data are needed so that the model can better balance efficiency and recognition performance in practical wearable deployments.

C. Combined Analysis of Real and Synthetic Datasets

To address the performance degradation observed under low-sampling-rate conditions, where each 5-s window contains only 20 samples, we investigate whether augmenting the training data with synthetic sequences can improve model generalization. In particular, we evaluate a hybrid training strategy that combines real inertial recordings with synthetically generated sequences and assess its impact on both overall accuracy and class-wise performance. Since the right-wrist dataset is available from multiple subjects and, after downsampling, provides up to 300 samples per class, it offers sufficient coverage for model training. In contrast, the left-wrist dataset is available only from a single subject and is therefore more susceptible to data scarcity. For this reason, the left-wrist data are used as the basis for synthetic sample generation to enrich the limited training set. This hybrid formulation yields a more balanced dataset by integrating real data from both wrists, multisubject recordings, and synthetic samples, thereby improving diversity and supporting broader generalization.

1) *Synthetic Data Generation via Conditional WGAN-GP*: To mitigate data scarcity in the real pitch-roll dataset acquired from human subjects, a conditional Wasserstein generative

TABLE I
CONDITIONAL WGAN-GP CONFIGURATION AND HYPERPARAMETERS

Parameter	Value
<i>Data and preprocessing</i>	
GAN training dataset	<i>Dataset-S01.LW</i> (left wrist)
Gesture classes ($ \mathcal{Y} $)	4 (<i>arm, hand, hair, mouth</i>)
Sequence length (T)	20
Channels	2 (Pitch, Roll)
Real samples per class (GAN training)	100
Synthetic samples per class (generated)	100
Total samples per class (downstream)	200
Normalization	Min-Max scaling to $[0, 1]$
<i>Generator $G(\mathbf{z}, y)$</i>	
Latent dimension ($\mathbf{z}$)	128
Label embedding dimension ($\mathbf{e}_y$)	32
Generator input dimension ($\mathbf{h}_0$)	160 (=128+32)
MLP architecture	FC(160→256→512→1024→40)
Hidden / output activations	ReLU / Sigmoid
Dropout probability p	0.2
Output tensor shape	(1, 20, 2)
Weight initialization	Xavier uniform (gain=1.0); embed $\sim \mathcal{N}(0, 0.02)$
<i>Critic $D(\mathbf{x}, y)$</i>	
Critic input tensor shape	(1, 20, 2)
Conv backbone (kernel)	1→64→128→256 (3×3)
Stride / padding	$s=1, p=1$; $s=2, p=1$; $s=2, p=1$
Activation	LeakyReLU ($\alpha = 0.2$)
Pooling	AdaptiveAvgPool2d(1, 1)
Feature fusion	256D features + 32D label embed (=288D)
Final MLP	FC(288→128→1)
Dropout probability p	0.2
Weight initialization	Xavier uniform; embed $\sim \mathcal{N}(0, 0.02)$
<i>Optimization</i>	
Epochs	1000
Batch size	32
Critic updates per generator update (N_{critic})	4
Gradient penalty weight (λ_{GP})	10
Gradient clipping (max norm)	1.0
Optimizer	Adam
Learning rates (η_D, η_G)	$3 \times 10^{-4}, 3 \times 10^{-4}$
Adam parameters (β_1, β_2)	0.0, 0.9

adversarial network (GAN) with gradient penalty (cWGAN-GP) is employed to synthesize class-conditioned time-series sequences for data augmentation. cWGAN-GP is preferred over conventional augmentation and standard conditional GANs for three reasons. First, simple augmentations (e.g., scaling or noise injection) may not preserve gesture-specific temporal dynamics in pitch-roll trajectories and can distort physically plausible motion patterns. Second, standard conditional GANs often exhibit training instability and mode collapse, which reduces synthetic diversity when gradient signals vanish under limited distributional overlap. Third, cWGAN-GP minimizes the Wasserstein distance, providing stable and informative gradients throughout training, while the gradient penalty enforces the Lipschitz constraint without weight clipping. Conditioning on the class label further encourages preservation of class-specific statistics and temporal structure, which is later confirmed via distributional and statistical agreement between real and synthetic samples.

Synthetic samples are generated using *Dataset-S01.LW* (left-wrist recordings), where each motion segment is represented as $\mathbf{x} \in \mathbb{R}^{T \times 2}$ with two channels (pitch and roll) and a fixed temporal length of T samples. Each segment is associated with a gesture label $y \in \{0, 1, 2, 3\}$ corresponding to *arm, hand, hair, mouth*, and generation is driven by a latent vector $\mathbf{z} \in \mathbb{R}^{d_z}$ sampled from $\mathcal{N}(\mathbf{0}, \mathbf{I})$. Fig. 8(a) summarizes the conditional generation and critic-based training workflow. All available real samples from the four gesture classes are used to train the cWGAN-GP to maximize the effective sample size for distribution learning. For each class, the recorded sequences

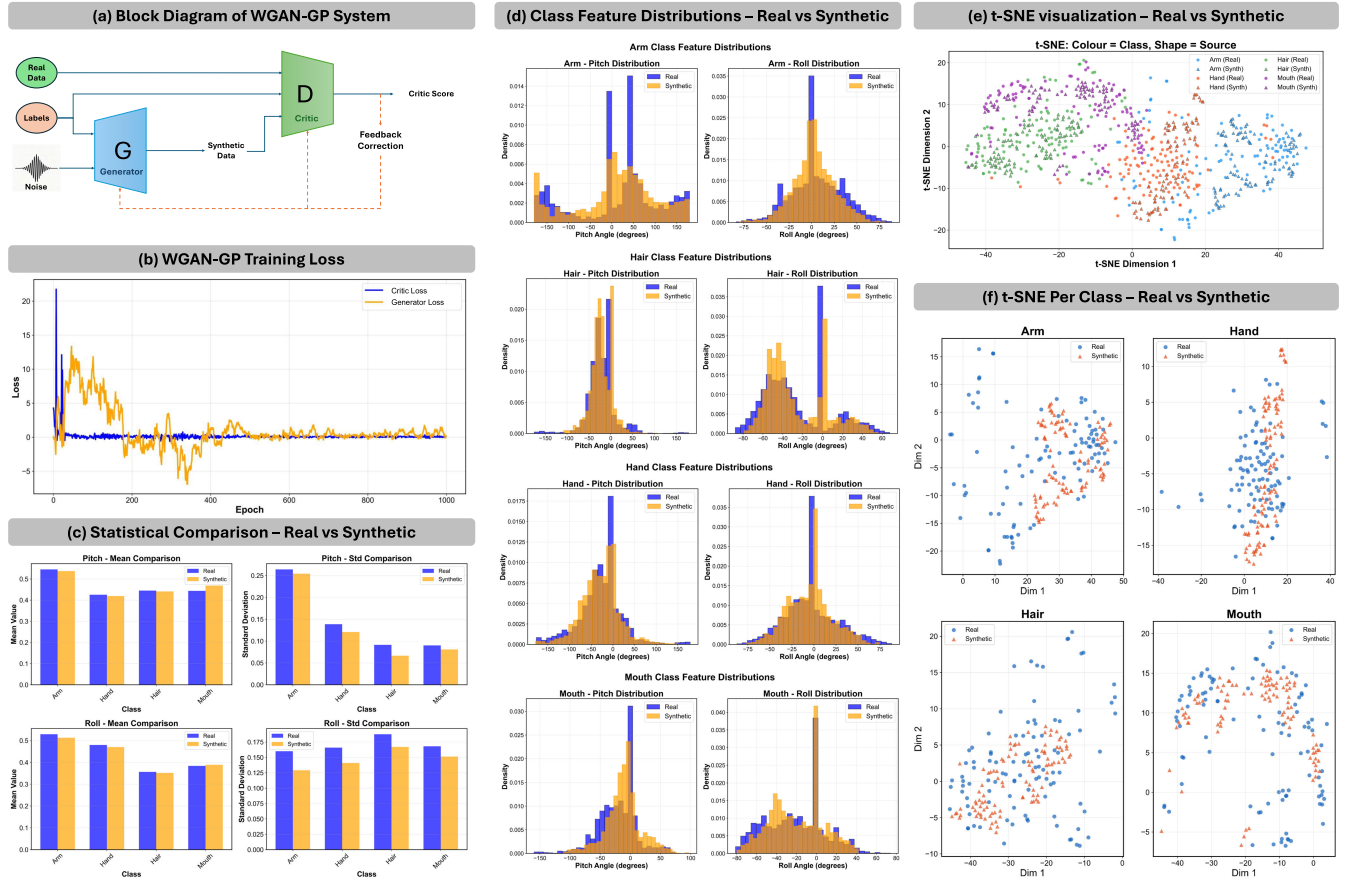


Fig. 8. Conditional WGAN-GP-based synthetic data generation and validation. (a) Class-conditioned generator–critic architecture for generating pitch–roll sequences from latent noise and gesture labels. (b) Generator and critic loss curves during training. (c) Mean and standard deviation of pitch and roll for real and synthetic samples. (d) Real and generated pitch–roll distributions across gesture classes. (e) t-SNE visualization of the complete dataset, with colors indicating classes and markers distinguishing real and synthetic samples. (f) Class-wise t-SNE distributions.

are parsed and formatted into fixed-length tensors of size $T \times 2$ (via truncation or zero-padding) to enable batch training. The dataset is normalized using Min–Max scaling to $[0, 1]$ across both channels, consistent with a sigmoid generator output. The complete preprocessing pipeline, network configurations, and training hyperparameters are reported in Table I.

The conditional generator $G(\mathbf{z}, y)$ takes as input a latent noise vector $\mathbf{z} \in \mathbb{R}^{d_z}$ and a discrete class label $y \in \mathcal{Y} = \{0, \dots, |\mathcal{Y}| - 1\}$. The label y is mapped to a learnable class-embedding vector $\mathbf{e}_y \in \mathbb{R}^{d_e}$ using an embedding layer, and the generator input is formed by concatenation $\mathbf{h}_0 = [\mathbf{z}; \mathbf{e}_y] \in \mathbb{R}^{d_z + d_e}$. The vector $\mathbf{h}_0$ is then passed through an MLP composed of FC layers with ReLU activations and dropout with probability p , producing an output vector $\mathbf{g} \in [0, 1]^{2T}$. Finally, $\mathbf{g}$ is reshaped into a synthetic time-series segment $\tilde{\mathbf{x}} = G(\mathbf{z}, y) \in [0, 1]^{T \times 2}$, where the two channels correspond to pitch and roll; a sigmoid output activation ensures consistency with the Min–Max normalization applied to the training data.

The conditional critic $D(\mathbf{x}, y)$ assigns a real-valued score to an input sequence $\mathbf{x} \in \mathbb{R}^{T \times 2}$ under label conditioning y , and it is trained to approximate the Wasserstein-1 distance between the real conditional distribution $p_{\text{data}}(\mathbf{x} | y)$ and the model distribution induced by $G(\mathbf{z}, y)$. Let $\mathbf{x} \sim p_{\text{data}}$ denote real samples, $\mathbf{z} \sim p(\mathbf{z})$ [typically $\mathcal{N}(\mathbf{0}, \mathbf{I})$] denote latent noise, and $\tilde{\mathbf{x}} = G(\mathbf{z}, y)$ denote generated samples. To enforce the

1-Lipschitz constraint via the gradient penalty, interpolated samples are constructed as $\tilde{\mathbf{x}} = \epsilon \mathbf{x} + (1 - \epsilon)\tilde{\mathbf{x}}$, where $\epsilon \sim \mathcal{U}(0, 1)$, and the critic is optimized with the WGAN-GP objective (Wasserstein loss plus a gradient-penalty term weighted by λ_{GP}). The critic loss is

$$\mathcal{L}_D = \mathbb{E}_{\tilde{\mathbf{x}}, y} [D(\tilde{\mathbf{x}}, y)] - \mathbb{E}_{\mathbf{x}, y} [D(\mathbf{x}, y)] + \lambda_{\text{GP}} \mathbb{E}_{\tilde{\mathbf{x}}, y} (\|\nabla_{\tilde{\mathbf{x}}} D(\tilde{\mathbf{x}}, y)\|_2 - 1)^2. \quad (3)$$

The generator is optimized to maximize the critic score on synthetic samples, equivalently minimizing

$$\mathcal{L}_G = -\mathbb{E}_{\tilde{\mathbf{x}}, y} [D(\tilde{\mathbf{x}}, y)]. \quad (4)$$

In each iteration, the critic is updated multiple times per generator update to maintain a well-trained critic during early training. The training dynamics in Fig. 8(b) indicate that both critic and generator losses stabilize after approximately 500 epochs. After convergence, 100 synthetic sequences per class are generated by sampling $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and selecting the target label y ; samples are then inverse-transformed using the fit Min–Max scaler to recover pitch and roll values in degrees before being saved for downstream use. Random-motion segments are excluded from GAN training because they do not follow a consistent gesture-dependent pattern; instead, these samples are recollected from real participants.

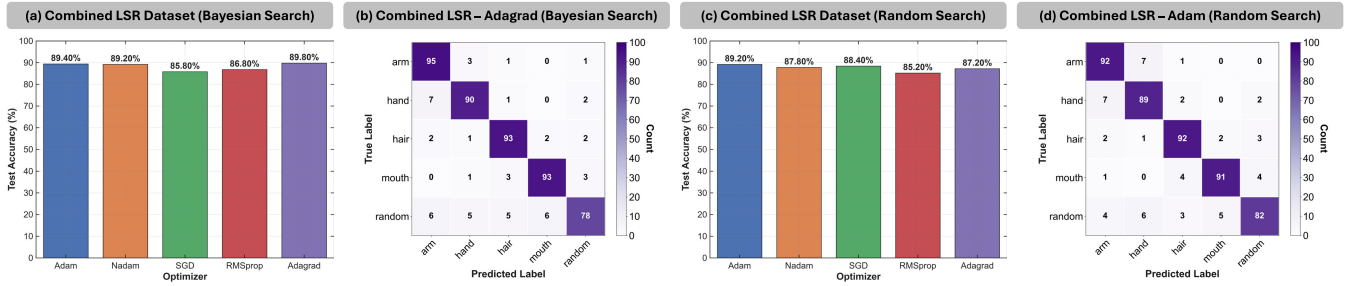


Fig. 9. Classification performance of the proposed 1-D CNN under low-sampling-rate inertial sensing with real and augmented data. (a) Optimizer-wise test accuracy using Bayesian hyperparameter optimization on the combined dataset. (b) Confusion matrix of the best-performing Adagrad model under Bayesian search. (c) Optimizer-wise test accuracy using random search on the combined dataset. (d) Confusion matrix of the best-performing Adam model under random search.

Consequently, the downstream dataset contains 200 samples per gesture class (100 real + 100 synthetic).

Beyond loss-based monitoring, the similarity between real and synthetic data is further assessed using per-class feature distributions and summary statistics. Fig. 8(c) compares the mean and standard deviation of pitch and roll for real and synthetic samples, providing a compact statistical validation of the generated data quality across classes. Fig. 8(d) shows substantial overlap between the real and generated pitch and roll distributions across the four gesture classes, indicating good marginal-level alignment. To complement these quantitative results, t-distributed stochastic neighbor embedding (t-SNE) analysis provides a qualitative view of the data structure in a 2-D space by measuring similarity relationships among points in the original high-dimensional space and mapping them into two dimensions while preserving local neighborhood structure as faithfully as possible. Fig. 8(e) presents the t-SNE visualization for the complete dataset, with different colors representing classes and different markers distinguishing real and synthetic samples, while Fig. 8(f) shows the class-wise t-SNE distributions. Together, these results confirm that the synthetic samples closely resemble the real data in both distribution and structure.

2) Classification Results on Combined Dataset: To evaluate the impact of synthetic data augmentation on classification performance, the proposed 1-D CNN is trained and tested on an augmented low-sampling-rate dataset that combines real inertial recordings with cWGAN-GP-generated synthetic sequences. The composite training set integrates four sources: downsampled right-wrist data from the single-subject recordings (100 sequences per class), newly collected low-sampling-rate left-wrist recordings from the single subject (100 sequences per class), downsampled right-wrist data from the multisubject recordings (200 sequences per class), and synthetic sequences generated at the same sampling rate using the left-wrist data (100 sequences per class). After integration, the resulting dataset contains 500 sequences per class, for a total of 2500 samples. The CNN is trained on this normalized composite set, which preserves class balance while increasing training diversity across multiple subjects, sampling configurations, and wrist orientations.

Fig. 9 summarizes the optimization results obtained with Bayesian search and random search, respectively. Under

Bayesian optimization, as shown in Fig. 9(a), Adagrad achieves the highest test accuracy of 89.80%, followed closely by Adam at 89.40% and Nadam at 89.20%, while SGD and RMSprop achieve 85.80% and 86.80%, respectively. The corresponding confusion matrix in Fig. 9(b) shows strong diagonal dominance, indicating reliable class separation across the five behaviors, with most errors concentrated in the random-motion class. Under random search, as shown in Fig. 9(c), Adam achieves the best performance with 89.20%, followed by Nadam at 88.40%, Adagrad at 87.20%, SGD at 88.40%, and RMSprop at 85.20%. The associated confusion matrix in Fig. 9(d) similarly exhibits a strong diagonal structure, confirming consistent discrimination across classes.

Compared with the datasets analyzed earlier, the combined dataset achieves comparable performance, indicating that the larger and more diverse training pool compensates for the reduced temporal resolution. This suggests that the addition of real and synthetic samples increases the effective coverage of motion variability, making the model more robust to intersubject differences, wrist orientation changes, and low-rate sampling effects.

V. DISCUSSION, LIMITATIONS, AND FUTURE WORK

As summarized in Table II, prior wearable systems for developmental disorder (DD) behavior monitoring have demonstrated promising performance; however, direct comparison across studies remains challenging due to substantial variability in target behaviors, sensing modalities, experimental protocols, subject populations, and annotation methodologies. Many existing approaches rely on commercially available devices not specifically designed for DD monitoring, and their evaluations are often constrained by limited sample sizes, reuse of public datasets, or reliance on video-based labeling. A key limitation in the literature is the difficulty of capturing authentic stereotypic behaviors under natural conditions. In several studies, target behaviors are elicited through structured prompting rather than observed spontaneously, which limits the ability to represent the inherent variability and contextual dependence of real-world behavior. This gap underscores the need for wearable systems validated under ecologically realistic conditions. Recent work by Caroppo et al. [44] further emphasizes the increasing interest in smartwatch-based

TABLE II
COMPARISON OF WEARABLE SYSTEMS FOR REPETITIVE BEHAVIOR MONITORING IN DDs

Work	Behavior coverage	Sensor burden	Device / platform	Sampling rate	Evaluation setting	Accuracy	User burden	Key remarks
2011 [33]	Hand flapping, body rocking	3 accelerometers (wrists + torso)	MITes wireless accelerometer	60 Hz	Laboratory + classroom	~90%	Moderate to high; multi-sensor setup	Video-validated early wearable approach; 6 ASD individuals; start and end times of movements annotated from video.
2013 [34]	Hand flapping	1 accelerometer (wrist)	TI EZ430-Chronos	80 Hz	School	~82%	Low; single-device wearable	Single-wrist wearable approach; 2 ASD individuals; compared with video analysis and trained/tested on non-ASD participants.
2014 [35]	Hand flapping, body rocking	3 accelerometers (wrists + torso)	MITes / Wockets	60/90 Hz	Classroom	81–99%	Moderate to high; multi-node system	Multi-sensor wearable system; 6 ASD individuals; video annotation used and variability across subjects observed.
2017 [36]	Arm-related actions	IMU (wrist)	Sony SWR50	20 Hz	Summer camp	77–91%	Low; smartwatch form factor	Smartwatch-based arm-action detection; 7 individuals with DD; timing reference obtained through video surveillance.
2017 [37]	Flapping, painting, sipping	Accelerometer (wrist)	Moto 360	50 Hz	Imitation + Naturalistic	~94%	Low; consumer smartwatch	Consumer smartwatch feasibility study; 12 healthy subjects and 2 ASD subjects; demonstrated wrist-worn deployment.
2017 [38]	Hand flapping, body rocking	3 accelerometers (wrists + torso)	MITes / Wockets	60/90 Hz	Classroom	>86%	Moderate to high; multi-sensor system	Public-dataset evaluation with multi-sensor setup; 6 ASD individuals; classroom-based assessment.
2018 [39]	Hand flapping, head banging, dropping	IMU (wrist)	Moto 360	32/16 Hz	Controlled imitation	~92%	Low; smartwatch-based	Commercial smartwatch study; simulated data from adults imitating children with ASD; naturalistic scenario considered.
2018 [25]	Hand flapping, body rocking	4 accelerometers (wrists + torso)	Zephyr BioPatch	100 Hz	Laboratory	>80%	High; sensor-heavy	Sensor-heavy wearable configuration; ASD individuals involved; video coding used for labeling.
2018 [40]	SIB, emotional dysregulation, aggression	EDA + HR + accelerometer	Empatica E4	–	Naturalistic	>80%	Moderate to high; multi-modal wearable	Multimodal physiological monitoring; 70 individuals with DD; aggressive behavior predicted up to 3 minutes in advance.
2021 [41]	Gesture recognition	IMU (wrist)	Hexiwear sensors module	50 Hz	Controlled	~91%	Low; single wearable node	Single wearable-node gesture study; 10 ASD individuals; 24 gestures evaluated.
2024 [42]	Hand flapping, rocking	5 IMUs + bio-signals	Custom system + Empatica E4	64 Hz phys., 441 kHz audio	Controlled	~88%	High; highly instrumented	Highly instrumented custom system; individual-specific models used; lab-based multimodal validation performed.
2025 [43]	Hand flapping, rocking	IMU (wrist)	Custom IMU setup	100 Hz	Controlled	>90%	High; computation-intensive	Multi-IMU deep-learning system; 41 individuals with DD; controlled evaluation with high computational demand.
Proposed	Hand/arm flapping, hair pulling/twisting, mouthing	2 accelerometers	Custom low-power HBC setup	26/4 Hz	Controlled imitation + augmented	>89%	Low; smartwatch form factor	Low-power HBC-based system; designed for unobtrusive naturalistic monitoring; evaluated on data from 20 subjects

DD: developmental disorder, HBC: human body communication.

solutions for movement disorder monitoring and highlights the importance of practical, deployable platforms.

Within this context, the proposed system offers several distinct advantages. It focuses on common upper limb stereotypic behaviors using a wrist-worn, smartwatch-form-factor device that supports unobtrusive and practical deployment. To the best of our knowledge, this is the first study to utilize HBC as an intrabody communication mechanism for this application, enabling a low-power architecture suitable for long-term monitoring. In addition, the system operates at substantially lower sampling rates compared to many existing approaches while maintaining competitive classification performance, demonstrating an effective tradeoff between computational efficiency and accuracy. Experimental results demonstrate a consistent performance hierarchy across both high- and low-rate datasets, with the temporal 1-D CNN outperforming conventional ML and MLP-based models. The evaluation employs a randomly partitioned train/test split, with careful verification to prevent data leakage, thereby reflecting true cross-subject generalization rather than memorization

of participant-specific characteristics. These results highlight the effectiveness of convolutional temporal feature extraction for robust recognition from pitch-roll kinematic sequences, supporting the practicality and efficiency of the proposed framework for wearable behavior recognition.

Despite these contributions, several limitations remain. The current framework employs fixed-duration 5-s segmentation and is not yet optimized for continuous real-time inference on streaming data. Future work will focus on enabling streaming-based segmentation, online inference, and temporal postprocessing to support detection of behavior onset, offset, duration, and frequency in naturalistic settings. Although the wearable is designed for a fixed wrist orientation, data collection from both arms was incorporated to improve robustness to placement variability; however, the current system does not explicitly differentiate between left- and right-arm movements. Future investigations will explore orientation-aware sensing strategies, including the integration of gyroscope data, to enhance robustness under varying wearing conditions. Furthermore, the current evaluation relies on sample-level

splitting rather than a subject-independent protocol. Future work will incorporate a leave-one-subject-out evaluation scheme to better assess cross-subject generalization. Finally, validation has thus far been conducted outside the target clinical population. Evaluation involving individuals with ASD or other DD is a critical next step toward clinical translation, enabling more comprehensive assessment of usability, generalization, and real-world deployment potential.

VI. CONCLUSION

This study validates the feasibility of an HBC-enabled wearable sensor network for low-power and unobtrusive monitoring of upper limb repetitive behaviors in individuals with ASD. By utilizing capacitive-coupling HBC for intrabody communication among wrist-worn accelerometers, the proposed system achieves significantly higher energy efficiency and interference immunity than conventional RF-based methods, while maintaining signal fidelity comparable to direct measurements. Prototype evaluations and power-scaling analysis confirm that the custom 65-nm CMOS HBC TX IC, supporting OOK modulation, operates at only 9 μ A under nominal conditions, substantially lower than typical RF transceivers. On the behavioral classification front, Bayesian- and random-search optimized 1-D CNN models attain 91% accuracy on real, single-subject high-sampling-rate data of 26 Hz. Extending this to multisubjects, the optimized CNN achieves an accuracy of 89.5%. In the class-imbalanced scenario where random movements occur more frequently, the CNN models classify them effectively, enhancing accuracy to 91.25%. Moreover, augmenting the collected dataset at a low sampling rate of 4 Hz with cWGAN-generated synthetic sequences increases data diversity and results in 89.8% classification performance across hand/arm flapping, hair twisting, and mouthing behaviors. These findings underscore HBC-based sensing combined with real-synthetic data augmentation as a scalable and energy-efficient approach for continuous, naturalistic behavioral monitoring, paving the way for earlier detection and adaptive intervention in ASD care.

ACKNOWLEDGMENT

The authors would like to thank Dr. Paul O'Callaghan and Dr. Zainab Aldawood from the Family and Child Support Center (FCSC) Team, King Abdullah University of Science and Technology, for their valuable guidance and insights regarding autism-related behaviors.

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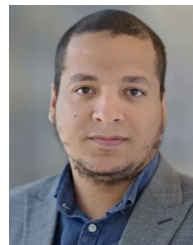
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